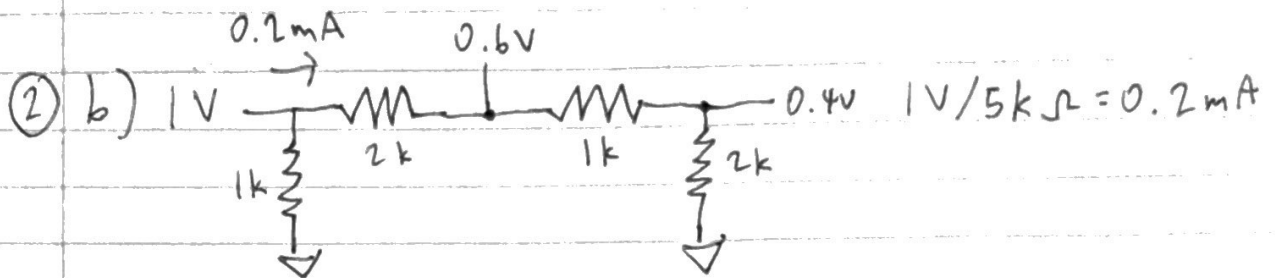


$$\textcircled{1} I_{\text{col}0} = 1\text{V} / 1\text{k}\Omega = \boxed{1\text{mA}}$$



$$1\text{V} - (0.2\text{mA})(2\text{k}\Omega) = \boxed{0.6\text{V} = V_{\text{col}1}}$$

$$1\text{V} - 0.4\text{V} - (0.2\text{mA})(1\text{k}) = \boxed{0.4\text{V} = V_{\text{row}1}}$$

$$\text{c) } I_{\text{col}0} = I_{\text{direct}} + I_{\text{sneak}}$$

$$= (V_{\text{row}0} - V_{\text{col}0}) / 1\text{k}\Omega + (V_{\text{row}1} - V_{\text{col}0}) / 2\text{k}\Omega$$

$$= (1\text{V} - 0\text{V}) / 1\text{k} + (0.4\text{V} - 0\text{V}) / 2\text{k}$$

$$= 1\text{mA} + 0.2\text{mA}$$

$$= \boxed{1.2\text{mA}}$$

③ I believe it's because of the floating rows and columns, and that causes leaks for the current. For large crossbar arrays, having a lot of these sneak paths could affect the current/signal.